

OBJECT ORIENTED PROGRAMMING

LAB ASSIGNMENT-2

(Topics – Constructors and Destructors)

Programming Questions

1. Write a C++ program to create a class **Student** with data members **name** and **rollNo**. Use a **default constructor** to initialize these values and display the student details.
2. Create a class Rectangle having data members length and breadth. Use a **parameterized constructor** to initialize the values and display the area of the rectangle.
3. Write a C++ program to create a class Employee with data members name, id, and salary. Initialize the data members using a parameterized constructor and display the employee details.
4. Create a class Box with data members length, width, and height. Implement:
 - a default constructor,
 - a parameterized constructor,
 - a function to calculate and display volume.
5. Write a program to demonstrate **constructor overloading** using a class Student. Create:
 - a default constructor,
 - a constructor accepting name and roll number,
 - a constructor accepting name, roll number, and CGPA.Display the details of objects created using all three constructors.
6. Create a class Complex with real and imaginary data members. Implement:
 - a default constructor,
 - a parameterized constructor,
 - a display function.
7. Create a class Demo that displays a message when its constructor is called and another message when its destructor is called. Create multiple objects of the class and observe the order in which constructors and destructors execute.
8. Write a C++ program to demonstrate the **execution order of constructors and destructors** when objects are created inside main() and inside a separate function.